# Project Report

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Link to Colab Notebook

## Project Description

Medical misinformation threatens public health by deterring people from seeking proper medical care. The goal of this project is to build an automated classification model that accurately labels short-form medical claims as factual or fabricated. To achieve this goal, traditional machine learning is not ideal because it ignores linguistic context and word order. Misinformation is often deceptive, and a model that relies on keyword frequency will not be able to classify such text accurately. Deep learning is required because it processes text sequentially, enabling it to learn the contextual and grammatical patterns of deception rather than merely following words (1).

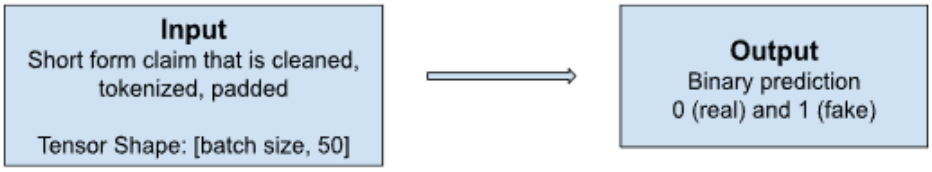



Figure 1: Model takes tokenized short text input and outputs real (0) or fake (1).

## Data Processing

Medical claims from CoAID and other misinformation from FakeNewsPrediction were collected and combined into a unified dataset (2)(3). The datasets did not have the same format which made it challenging to resolve. For example, CoAID did not have a dedicated data column for 'Real/Fake' labels and instead put that information within the claim names. The raw text was then cleaned by removing special characters and standardized casing, tokenized to a dynamic vocabulary, and padded to a 50-word sequence for uniform tensor dimensions. There are 6362 claims in the final dataset, with 3171 being real and 3191 false. The claims were then split into a 70% training set, 15% validation set, and 15% testing set.

| | |
|---|---|
| **Original Text** | How Hillary Clinton Could Win |
| **Cleaned & Tokenized** | ['hillary', 'clinton', 'could', 'win'] |
| **Label** | 0 |

Figure 2: Sample of unclean and clean data, showcasing data format (text, label).

Aside from the 15% of the data that is set aside for testing, new and unseen data will also be collected from recent tweets for the ultimate test.

## Baseline Model

A simple baseline model was created to compare to the primary model. The raw text was processed using a TF-IDF Vectorizer and passed into a Logistic Regression classifier (4). For each word in the input text, the model will assign a decimal at the corresponding index. The model will count how often specific keywords appear in a post, ignoring word order and context, making it a good baseline for comparison.

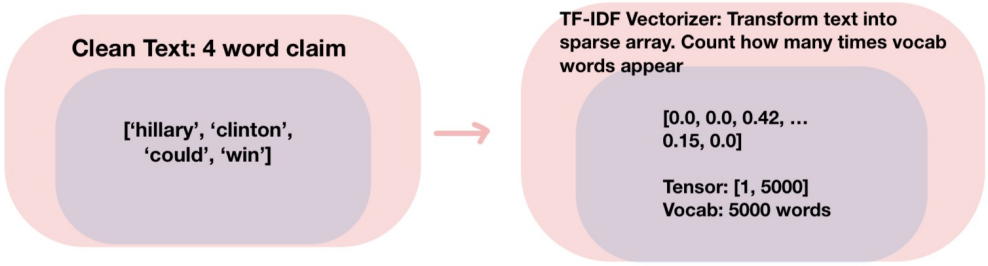



Figure 3: Flowchart showing text to TF-IDF Vectorizer to Logistic Regression Output.

The model achieved an initial accuracy of 79.37% and flagged several words as misinformation or real claims. Some of the words flagged as misinformation are "wikileaks", "breaking", "fbi", and "comment". The results show that the dataset has distinct linguistic patterns and confirm that the project is feasible. However, the baseline model fails to detect misinformation that rely on sentence structure rather than specific trigger words, which is a hurdle that needs to be improved for the primary model.

## Primary Model

The proposed architecture is a custom MisinfoLSTM that allows the model to learn word order and context. First, the embedding layer will convert the 50-word tokenized input into semantic vectors.




Figure 4: Pipeline of raw text tokenization, padding, and embedding vectorization.

The core model is a 3-layer bidirectional LSTM that will process the input forwards and backwards to capture maximum context (5). The forward and backward passes produce separate tensors representing their respective contexts, which are then concatenated into a single tensor before going through the linear classification layer. A Sigmoid activation will serve as a classifier, outputting a probability between 0 (real) and 1 (fake).

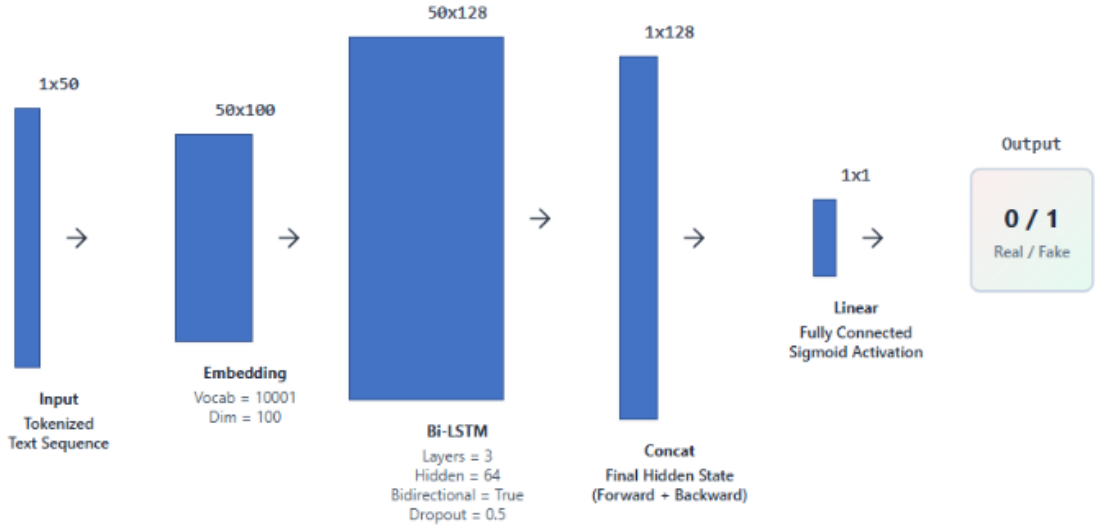



Figure 5: Architecture of the 3-Layer Bidirectional LSTM model.

The training loop utilized Binary Cross Entropy Loss to show the model's performance (6). The training error converged to near zero within 10 epochs, indicating the model's ability to learn and classify real and fake text. The current model uses the Adam optimizer with a 0.01 learning rate, 32 batch size, and 30 epochs. However, the results show spiking validation loss due to overfitting. The current model struggles to generalize, which is a challenge, and the highest accuracy achieved so far is only around 78%. The next steps of the project will focus on tuning the model through hyperparameter tuning, adjusting the custom MisinfoLSTM, and utilizing pre-trained models to improve the baseline accuracy and overall performance. A pre-trained model with a significantly larger vocabulary can provide the network with a pre-existing and deeper understanding of complex word relationships. Because the model no longer has to learn language patterns from a limited dataset, it can leverage this broader linguistic foundation to reduce overfitting and generalize more effectively.

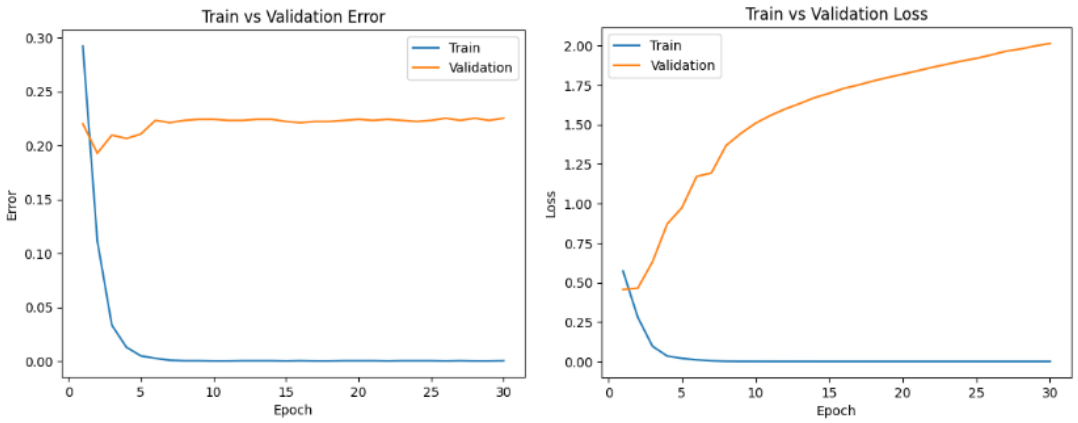



Figure 6: Train vs Validation Error and Loss showing overfitting after Epoch 3.

## References

(1) (2) (3) (4) (5) (6)

## References


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[2] Arashnic, "COVID-19 fake news dataset," 2020. [Online]. Available: https://www.kaggle.com/datasets/arashnic/covid19-fake-news/data

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